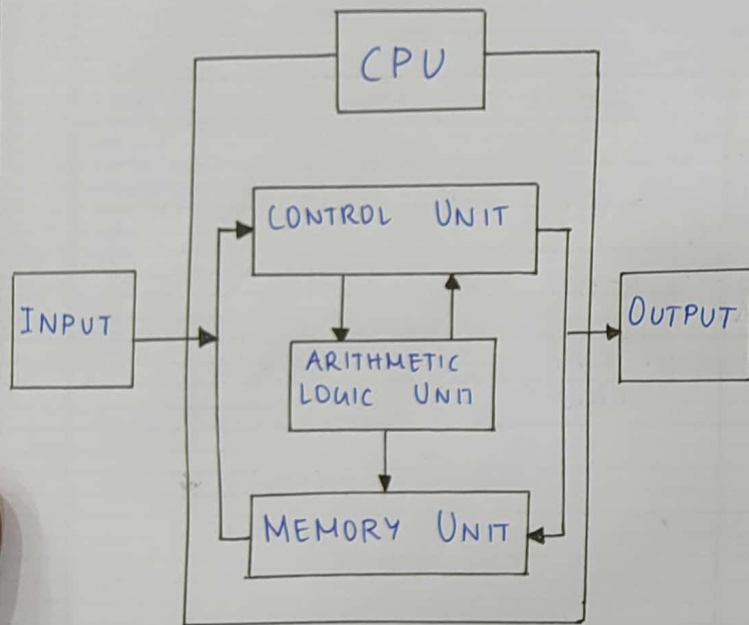


## BLOCK DIAGRAM OF COMPUTER



## Experiment - 1

### FAMILIARIZATION OF COMPUTER

#### AIM

To study the fundamentals of computer

#### COMPUTER

A computer is a programming machine which designs sequentially and automatically carries out a sequence of arithmetic & logic operations. The program of a computer can be changed allowing the person to solve more and more kinds of problems.

First electronic computer was developed in mid 20<sup>th</sup> century. Originally they were the size of large rooms consuming the space equivalent of 700 of modern computers.

Modern computers provide and perform a variety of calculation or computation in a time efficient manner.

Modern computers use digital data. This digital data is in the form of 0s representing ON & OFF status.

#### GENERATIONS OF COMPUTER

The 1<sup>st</sup> generation electronic computers used vacuum tubes. The computer ENIAC was the 1<sup>st</sup> general purpose electronic computer.

## TECHNOLOGY USED IN DIFFERENT GENERATIONS OF COMPUTERS

| GENERATION | TECHNOLOGY  | OPERATING SYSTEM | YEAR OF INTRODUCTION | SPECIFIC COMPUTERS             |
|------------|-------------|------------------|----------------------|--------------------------------|
| 1.         | Vacuum Tube | None             | 1945                 | Mark II                        |
| 2.         | Transistor  | None             | 1956                 | IBM 1401, ICL 1901, B5000      |
| 3.         | SSI & MSI   | Yes              | 1964                 | IBM 5/360/370, UNIVAC 1100     |
| 4.         | LSI & VLSI  | Yes              | 1971                 | ICL 2900, HP 9845A, VAX 11/780 |
| 5.         | HAL         | Yes              | Present & Beyond     |                                |

II<sup>nd</sup> generation computers used transistors instead of vacuum tubes. They were small, consumed less power and produced less heat.

III<sup>rd</sup> generation computers used IC. These increased computing power and reduced size.

IV<sup>th</sup> generation were based on microprocessor chips. This gave us even higher computational power.

### PARTS OF A COMPUTER

A computer system can be divided into 2 components which are responsible for providing the mechanisms to input & output data and to electrically control the various input, output and their storage. They are known as hardware and software.

### HARDWARE

It is the physical components of a computer that includes all mechanical electrical electronic and magnetic parts attached to it. The major hardware components are

1. Input output devices
2. Central processing unit
3. Memory unit & storage device
4. Interface unit



## 1. Input Output devices

### Input device

The data and instructions are typed, submitted or transmitted to a computer through input devices. Input devices are electronic or electromechanical equipment that provide a mean of communication with the computer system for feeding input data and instructions.

eg: Keyboard, mouse, scanner

### Output device

It mirrors the input data or shows the output results of the operations on the input data or print the data. The most common output devices are monitor. The printer is used to print the result.

## 2. CPU

Central processing unit (CPU) can be treated as the brain of the computer. Most of the processing takes place in CPU. During processing it locates and executes the program instructions. It also fetches data from memory & input or output devices and sends data back.

## 3. Memory Unit

Components such as input devices, output devices and CPU are not sufficient for making a

computer work. A storage area is needed in a computer to store instructions & data, either temporarily or permanently. So that subsequent retrieval of instructions and data can be possible on demand.

#### 4. Interface Unit

It interconnects CPU with memory and also with the various input/output devices. The instruction and data move between the CPU and other hardware components through interface unit.

#### SOFTWARE

They are set of programs and instructions. There are 2 types

1. System software
2. Application software

##### 1. System software

Set of programs which manage and control the internal operations of a computer system. It is a collection of programs written to service other programs.

##### 2. Application software

To instruct particular application. Stand alone programme that solve problems only.



in between is called source code. It is a human-readable code that is written in a high-level programming language. It is the code that the computer can understand and execute.

Source code is written in a text file. It is a plain text file that contains the instructions for the computer to follow. It is the code that the compiler uses to generate the executable file.

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## OPEN SOURCE SOFTWARE

It is software with source code that anyone can inspect, modify & enhance. "Source code" is the part of software that most computer users don't even see, it's the code, computer programs can manipulate to change. Now a piece of software a "program" or "application" - works. Programmers who have access to a computer program's source code can improve that program by adding features to it or fixing parts that always don't work correctly.

## BASIC STRUCTURE OF A C PROGRAM:

The 'C' Compiler :

The source code written in source file is the human readable source of your programme. It needs to be compiled into machine language so that CPU can actually execute the program as per the instruction given.

The compiler compiles the source code into final executable program. The most frequently used and free available compiler, otherwise you can have the respective operating system.

## Steps to Compile and Run a C program in UBUNTU

Use a ~~text~~ editor to create a 'C' source code

```
#include <stdio.h>
void main()
{
    printf("Hello World");
}
```

1. Save file as hello.c in desktop
2. Open the terminal and write the path of the file (here cd Desktop)
3. Compile the program with the command gcc hello.c  
The GNU compilers collection (gcc) converts a source file to executable file by preprocessing, compilation, assembly, linking
4. Execute the program with command ./a.out  
Hello World will be displayed on the screen

### RESULT

The fundamentals of computer are studied